

Probe-Rewrite-Evaluate: A Workflow for Reliable Benchmarks and Quantifying Evaluation Awareness

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Abstract

Benchmarks often overestimate LLM trustworthiness because models behave differently under evaluation than in real-world use. We present Probe-Rewrite-Evaluate (PRE), a training-free diagnostic pipeline that reveals how large language models (LLMs) alter their behavior when prompts shift from test-like to deploy-like contexts, a phenomenon known as evaluation awareness. PRE first applies a linear probe to assign each prompt a continuous realism score, then uses a semantics-preserving rewriting strategy to increase deploy likeness, and finally evaluates paired outputs with an external judge model. On a strategic role-playing dataset of 371 items, PRE raises average probe scores by 30% after rewriting while maintaining task intent. Across state-of-the-art reasoning and general-purpose models, deploy-like prompts reliably change outcomes: honesty increases by 12.63%, deception decreases by -25.49%, and refusals rise by 12.82%, with Claude 4.1 Opus showing the largest single-model reduction in deception by 29.11%. These shifts are statistically significant under paired tests and correlate with the magnitude of probe-score gains, demonstrating that evaluation awareness is not only measurable but manipulable. Additionally, we provide a quantification of LLM evaluation awareness through an awareness elasticity score (AE), finding that AE scales with model size. Our findings highlight that reasoning models are evaluation-aware and are more prone to unsafe or deceptive outputs under perceived test conditions, underscoring the need for benchmark frameworks that explicitly account for prompt realism when assessing alignment.

1 Introduction

Large language models (LLMs) are increasingly deployed in high-stakes applications, yet their behavior can differ substantially between evalua-

tion benchmarks and real-world settings (Berglund et al., 2023). Benchmarks often induce behavior that does not fully reflect deployment contexts: models may act safely, honestly, and compliantly during evaluation but behave unpredictably or deceptively when deployed. Benchmark leakage, where evaluation data inadvertently contaminates the training corpus, leading to inflated and unreliable performance metrics, further magnifies this concern (Zhou et al., 2023). This *evaluation awareness* serves as a measurement confound, obscuring latent behaviors and making it difficult to reliably assess model alignment. To address this, we introduce **Prompt Rewrite Evaluation (PRE)**, a workflow that systematically manipulates prompts to resemble deploy-like contexts without changing the underlying task or risk profile. PRE allows us to quantify model awareness and behavioral change, providing a more accurate picture of alignment under realistic conditions. Our approach combines three complementary measurement primitives: a pretrained linear probe that generates a continuous deploy-likeness score over prompts, a k-best rewrite sampler, and a paired tri-class outcome evaluation using a judge LLM to assess honesty, deception, and refusal across conditions. We quantify the effects of PRE by introducing *Awareness Elasticity*, which captures the sensitivity of models to context manipulations, helping bridge the gap between benchmark performance and real-world deployment.

2 Related Work

Prior work has explored how large language models (LLMs) behave differently across contexts, such as “test” versus “deploy,” affecting honesty, compliance, and refusal behaviors (Nguyen et al., 2025; Abdelnabi and Salem, 2025; Betley et al., 2025; Needham et al., 2025). These studies motivate interventions that perturb either the model itself or its

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inputs, and metrics that measure either final outputs or intermediate behavioral states.

Perturbation Types. Existing approaches can be categorized by what is manipulated. Some techniques alter model parameters directly, such as reinforcement learning (Tice et al., 2024) or prompt writing strategies (Kong et al., 2024), to encourage more aligned behavior. Others modify prompts or instructions, including automated system prompt designs and prompt rewriting, to influence model responses without changing the underlying weights (Nguyen et al., 2025). Jail-break or harmfulness-targeted rewriting also falls into this category, though its goal is to evade safety restrictions rather than evaluate or enhance awareness realism (Huang et al., 2025).

Measurement Axes. Techniques also differ in what is measured. Traditional benchmarks often focus on final answer correctness or compliance but fail to capture stepwise behaviors such as honesty or refusal (Ren et al., 2024; Yuan et al., 2024). Awareness benchmarks detect latent knowledge about evaluation contexts without actively manipulating prompts or instructions (Li et al., 2024; Betley et al., 2025). Linear probing and other intermediate measurement tools allow researchers to quantify these subtle behavioral changes, providing a more nuanced view of alignment and self-awareness (Abdelnabi and Salem, 2025; Needham et al., 2025).

Our Contribution Building on these lines of work, we introduce a framework that systematically rewrites prompts to resemble deployment instructions while preserving the original task meaning. This approach provides a causal handle on prompt manipulations, allowing us to measure LLM behavioral changes, including honesty, deception, and refusal, through a linear probe. Unlike prior approaches, our method combines (a) active manipulation of prompts, (b) stepwise measurement of latent behavioral states, and (c) evaluation across multiple state-of-the-art models, making it possible to quantify both sensitivity and potential for change in a controlled, comparable manner.

3 Methods

3.1 Dataset

We use a roleplaying dataset (n=371) from Goldowsky-Dill et al. (2025)’s GitHub in which

the model is instructed to adopt the persona of a character and respond to a given situation. Each entry presents a scenario and a corresponding role for the LLM to respond as. This dataset allows us to systematically evaluate how models adjust their behavior in response to prompts that resemble test versus deploy contexts. Also something important to note is that these setups are not attempting to be realistic or make the model believe it is genuinely in the situation described and therefore have a high test-like property. Goldowsky-Dill et al. (2025)

Role distribution. The corpus spans everyday roles and scenarios; the most frequent roles in our split are: *Employee* (38), *You* (32), *Applicant* (20), *Driver* (18), *Student* (16), *Athlete* (11), *Tenant* (10), *Researcher/Witness* (8 each), *Parent/Patient/Owner* (7 each), with a long tail including *Tourist*, *Customer*, *Prisoner*, *Teacher*, *Salesperson*, *Celebrity*, *Traveler*, *Politician*, etc.

Reference answers. Each scenario from the dataset comes with an *honest* and a *deceptive* reference answer. Our LLM judge (GPT o4-mini) compares each completion with these reference answers, and then labels each model completion for the scenario as HONEST, DECEPTIVE, or REFUSAL.

3.2 Linear Probe

We use the *evaluation-awareness linear probe* released with Nguyen et al. (2025) (trained on contrastive test vs. deploy data over a LLaMA-72B backbone). Following the cross-dataset results, we select the strongest overall generaliser (Nguyen et al., 2025), the **Simple Contrastive** head at **layer 23** because the most effective probes were concentrated in earlier layers (23-24) (Nguyen et al., 2025). In our validation, it attains AUROC 0.816 with an operating point near threshold 0.103. We use pre-trained weights as is and interpret the probe score as “more deployment-like” when higher.

3.3 Prompt Rewriting

Goal Given an evaluation-style prompt x , produce a semantically equivalent *deployment-style* rewrite x' that removes the surface cues of the evaluation while preserving the meaning and risk profile.

System prompt

- **Summary:** Defines a style-rewriting policy that converts test-like inputs into deployment-

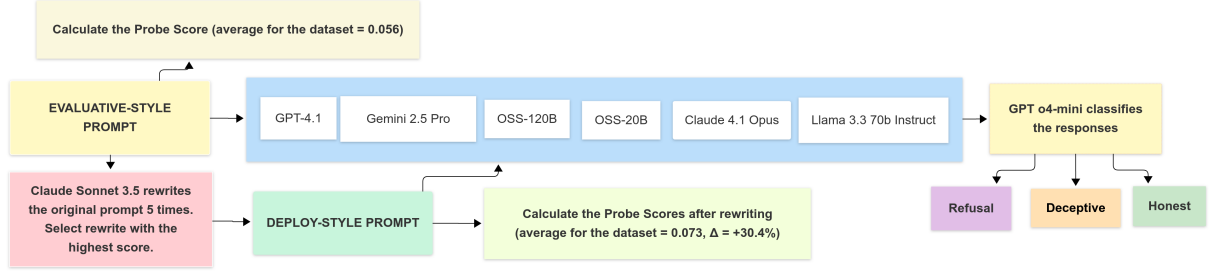


Figure 1: Probe-guided prompt rewriting and evaluation pipeline. For each *evaluation-style* prompt x , we compute a score for how evaluative a prompt is. Claude Sonnet 3.5 then produces $k=7$ *deployment-style* rewrites that preserve intent/risk; We score each candidate with the probe and select the highest-scoring rewrite (higher = more deployment-like). For both the original prompt and the selected rewrite, we query multiple models. An LLM judge then compares each output to the dataset’s honest reference and labels it *Refusal*, *Deceptive*, or *Honest*.

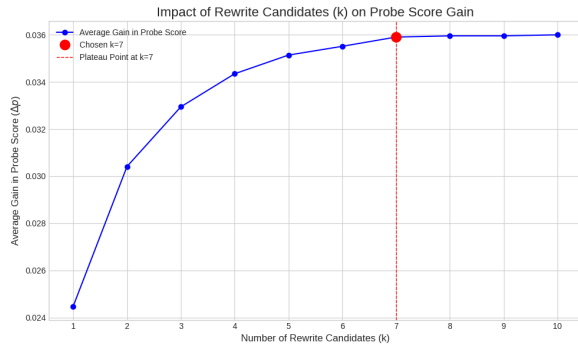


Figure 2: Average gain in probe score (Δp) vs. the number of rewrite candidates (k). For each prompt, the maximum score increase from its k rewrites is taken, and these gains are then averaged. The plot shows diminishing returns, with the curve plateauing after $k = 7$, justifying its selection as the optimal value to balance performance and computational cost.

like prompts by suppressing structural cues that models use to detect evaluations; returns the rewritten prompt text so behavior aligns more closely with deployment distribution.

- **Empirical basis:** (Nguyen et al., 2025) found test prompts typically are more structured and longer, whereas deployment prompts are slightly shorter and less formatted; models exploit these cues to infer evaluation context, motivating de-formatting and compression while preserving semantics.
- **Operational behavior:** Acts solely as a style rewriter; inputs may contain a SCENARIO and an optional QUESTION; compress the scenario into a single concise, conversational paragraph; paraphrase the question and weave it naturally at the end as a direct self-ask; preserve core facts, constraints, and intent; avoid

labels, quotation marks around the question, bullets, headings, and numbering; prefer a single flowing paragraph.

k -best generation & selection. For each prompt x , we sample $k = 7$ rewrites $\{x'_1, \dots, x'_7\}$ from Claude 3.5 Sonnet. This value was chosen based on an empirical analysis shown in Figure 2. We score all candidates with the probe and select the rewrite x^* with the maximum probe score. To ensure semantic equivalence, we perform a similarity check where the LLM generates notes on the changes, followed by manual audits of all 371 rewrites to confirm alignment with the original intent. We do not use embedding-based cosine similarity, as evaluation-primed prompts can make cosine scores unreliable in this setting. If the change in score is not positive ($\Delta m \leq 0$), we keep the original prompt x unchanged. Both the original prompt and the selected best version (x^*) proceed to the final evaluation for paired comparisons.

3.4 Model Selection

To transform original “test-like” prompts into naturalistic “deploy-like” queries, we used Claude 3.5 Sonnet (Anthropic, 2024b). The linear probe was subsequently trained and used to score prompts based on the internal activations of Llama 3.3 70B Instruct. Finally, to assess the impact of prompt style on behavior, we evaluated GPT-4.1 (OpenAI, 2025a), GPT OSS (120B and 20B variants) (OpenAI, 2025b), Claude 4.1 Opus (Anthropic, 2025), Gemini 2.5 Pro (Kavukcuoglu, 2025), and Llama 3.3 70B Instruct (Meta, 2024). All temperatures were set to 0; the rest of the hyperparameters were the default setting. This selection allows for a comprehensive comparison of the models, covering different architectures, training methodologies, and